

Pi-Lambda Theorem

Friday, September 11, 2015

Π - λ Theorem : If $\mathcal{C}$ is π -system, then $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$

Pf: $\lambda(\mathcal{C}) \subseteq \sigma(\mathcal{C})$ since $\sigma(\mathcal{C})$ is a λ -system ($\because \sigma$ -alg containing $\mathcal{C}$ and $\lambda(\mathcal{C})$ smallest such).

will show $\lambda(\mathcal{C})$ a σ -algebra. ($\Rightarrow \sigma(\mathcal{C}) \subset \lambda(\mathcal{C})$)

$\lambda(\mathcal{C})$: (i) $\Omega \in \lambda(\mathcal{C})$

(ii) $A \in \lambda(\mathcal{C}) \Rightarrow A^c = \Omega \setminus A \in \lambda(\mathcal{C})$.

(iii) $A_i \uparrow A_i \in \lambda(\mathcal{C}) = \cup A_i \in \lambda(\mathcal{C})$

only need to verify :

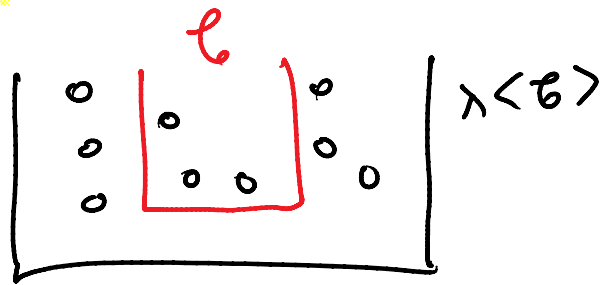
finite unions in $\lambda(\mathcal{C})$.

($\Rightarrow$) finite intersections in $\lambda(\mathcal{C})$

i.e will prove! $\forall A \in \lambda(\mathcal{C}), \forall B \in \lambda(\mathcal{C}) \Rightarrow A \cap B \in \lambda(\mathcal{C})$

1st prove ①^(c) if $A \in \lambda(\mathcal{C})$, if $B \in \mathcal{C} \Rightarrow A \cap B \in \lambda(\mathcal{C})$

1st prove ① if $A \in \lambda\langle \mathcal{C} \rangle$, if $B \in \mathcal{C} \Rightarrow A \cap B \in \lambda\langle \mathcal{C} \rangle$



define: (good set argument)

$$\lambda_1(\mathcal{C}) = \{ A \in \lambda\langle \mathcal{C} \rangle : \underline{A \cap B} \in \lambda\langle \mathcal{C} \rangle, \forall B \in \mathcal{C} \}$$

$$\textcircled{A} \subseteq \lambda\langle \mathcal{C} \rangle$$

(will show $\lambda_1(\mathcal{C}) = \lambda\langle \mathcal{C} \rangle \Rightarrow \textcircled{D} \text{ pvd}$)

$$\mathcal{C} \subseteq \lambda_1(\mathcal{C}) \quad (\text{since } A \in \mathcal{C}, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C} \subseteq \lambda\langle \mathcal{C} \rangle)$$

will prove $\lambda_1(\mathcal{C})$ is a λ -system.

(why? since $\lambda\langle \mathcal{C} \rangle$ smallest λ -syst $\supset \mathcal{C}$, $\lambda\langle \mathcal{C} \rangle \subseteq \lambda_1(\mathcal{C})$ $\textcircled{B}$)

$$\textcircled{A} + \textcircled{B} \Rightarrow \textcircled{C}$$

$$(i) \Omega \in \lambda_1(\mathcal{C})$$

$$(iii) A_i \uparrow \text{ in } \lambda_1(\mathcal{C})$$

$$\text{i.e. } \underline{A_i \uparrow}, A_i \in \lambda\langle \mathcal{C} \rangle, \underline{A_i \cap B} \in \lambda\langle \mathcal{C} \rangle \forall B \in \mathcal{C}.$$

$$\Rightarrow \bigcup A_i \in \lambda\langle \mathcal{C} \rangle \quad (\text{since } \lambda\langle \mathcal{C} \rangle \text{ } \lambda\text{-syst}) \quad (\bigcup A_i) \cap B$$

$$= \bigcup_i \underbrace{(A_i \cap B)}_{\uparrow \in \lambda\langle \mathcal{C} \rangle}$$

$$\Rightarrow \bigcup A_i \text{ in } \lambda_1(\mathcal{C})$$

$$\in \lambda(\mathcal{C})$$

(ii) Let $A, C \in \lambda_1(\mathcal{C})$ will check $(C \setminus A) \in \lambda_1(\mathcal{C})$

$$\underline{ACC}$$

$$A \in \lambda(\mathcal{C}), \quad A \cap B \in \lambda(\mathcal{C}) \quad \forall B \in \mathcal{C}.$$

$$C \in \lambda(\mathcal{C}), \quad C \cap B \in \lambda(\mathcal{C}) \quad \forall B \in \mathcal{C}.$$

$$\downarrow$$

$$(C \setminus A) \in \lambda(\mathcal{C})$$

$$\downarrow$$

$$(C \setminus A) \cap B = C \cap A^c \cap B$$

$$= (C \cap B) \cap (A \cap B)^c$$

$$= (C \cap B) \setminus (A \cap B) \in \lambda(\mathcal{C})$$

$$\text{Hence } C \setminus A \in \lambda_1(\mathcal{C})$$

$$\text{Hence } \lambda_1(\mathcal{C}) \text{ is } \lambda\text{-syst} \Rightarrow \lambda_1(\mathcal{C}) = \lambda(\mathcal{C})$$

i.e

$$\forall A \in \lambda(\mathcal{C}), \text{ and } \forall B \in \mathcal{C} \Rightarrow A \cap B \in \lambda(\mathcal{C})$$

$$* \left[\forall B \in \lambda\langle \mathcal{C} \rangle, \text{ and } \forall A \in \mathcal{C} \Rightarrow A \cap B \in \lambda\langle \mathcal{C} \rangle \right]$$

(2) ~~will prove~~ define. $\lambda_2(\mathcal{C}) = \left\{ \underbrace{A \in \lambda\langle \mathcal{C} \rangle}_{\subset \lambda\langle \mathcal{C} \rangle} : \begin{array}{l} A \cap B \in \lambda\langle \mathcal{C} \rangle \\ \forall B \in \lambda\langle \mathcal{C} \rangle \end{array} \right\}$

will prove: $\lambda_2(\mathcal{C}) = \lambda\langle \mathcal{C} \rangle$ (end of pf)

Note: $\mathcal{C} \subset \lambda_2(\mathcal{C})$ (from 1st step *)

($A \in \mathcal{C}$, from *) $A \in \lambda_2(\mathcal{C})$)

Now will show: $\lambda_2(\mathcal{C})$ is a λ -sys. ($\supset \mathcal{C}$)

(since $\mathcal{C} \subset \lambda_2(\mathcal{C})$, this will imply

$\lambda_2(\mathcal{C}) \supset \lambda\langle \mathcal{C} \rangle \Rightarrow$ end of pf)

$\rightarrow$ (pf is same/similar as 1st part pf.)